**********Section 1**********

Sample data to be loaded in Hive Tables

salesman.csv

```
5001,James Hoog,New York,0.15
5002,Nail Knite,Paris,0.13
5005,Pit Alex,London,0.11
5006,Mc Lyon,Paris,0.14
5003,Lauson, Hen,0.12
5007,Paul Adam,Rome,0.13
```

customer.csv

```
3002,Nick Rimando,New York,100,5001
3005,Graham Zusi,California,200,5002
3001,Brad Guzan,London,,5005
3004,Fabian Johns,Paris,300,5006
3007,Brad Davis,New York,200,5001
3009,Geoff Camero,Berlin,100,5003
3008,Julian Green,London,300,5002
3003,Jozy Altidor,Moscow,200,5007
```

orders.csv

```
70001,150.5,2012-10-05,3005,5002
70009,270.65,2012-09-10,3001,5005
70002,65.26,2012-10-05,3002,5001
70004,110.5,2012-08-17,3009,5003
70007,948.5,2012-09-10,3005,5002
70005,2400.6,2012-07-27,3007,5001
70008,5760,2012-09-10,3002,5001
70010,1983.43,2012-10-10,3004,5006
```

```
70003,2480.4,2012-10-10,3009,5003
70012,250.45,2012-06-27,3008,5002
70011,75.29,2012-08-17,3003,5007
70013,3045.6,2012-04-25,3002,5001
```

Hive Tables Metadata

salesman

```
salesman_id int,
name    string,
city      string,
commission double
```

customer

```
customer_id  int,
cust_name   string,
city    string,
grade  int,
 salesman_id int
```

orders

```
ord_no  int,
purch_amt  double,
ord_date  date,
customer_id int,
salesman_id int
```

Create a database named hive_test and create three tables

salesman

customer

orders

Create above tables with the help of given data and metadata.

**********Write SQL For Followings**********

- Write a SQL statement to prepare a list with salesman name, customer name and their cities for the salesmen and customer who belongs to the same city.

```
hive (hive_test)> SELECT
                >   s.name       AS salesman_name,
                >   c.cust_name  AS customer_name,
                >   s.city       AS city
                > FROM salesman s
                > JOIN customer c
                >   ON s.city = c.city;
```

- Write a SQL statement to know which salesman are working for which customer.

```
hive (hive_test)> SELECT
                >   s.name      AS salesman_name,
                >   c.cust_name AS customer_name
                > FROM customer c
                > JOIN salesman s
                >   ON c.salesman_id = s.salesman_id;
```

- Write a SQL statement to make a list with order no, purchase amount, customer name and their cities for those orders which order amount between 500 and 2000.

```
hive (hive_test)> SELECT
                >   o.ord_no,
                >   o.purch_amt,
                >   c.cust_name,
                >   c.city
                > FROM orders o
                > JOIN customer c
                >   ON o.customer_id = c.customer_id
                > WHERE o.purch_amt BETWEEN 500 AND 2000;
```

- Write a SQL statement to find the list of customers who appointed a salesman for their jobs who gets a commission from the company is more than 12%.

```
hive (hive_test)> SELECT DISTINCT
                >   c.customer_id,
                >   c.cust_name,
                >   c.city,
                >   s.name AS salesman_name,
                >   s.commission
                > FROM customer c
                > JOIN salesman s
                >   ON c.salesman_id = s.salesman_id
                > WHERE s.commission > 0.12;
```

- Write a SQL statement to find the list of customers who appointed a salesman for their jobs who does not live in the same city where their customer lives, and gets a commission above 12% .

```
hive (hive_test)> SELECT DISTINCT
                >   c.customer_id,
                >   c.cust_name,
                >   c.city AS customer_city,
                >   s.name AS salesman_name,
                >   s.city AS salesman_city,
                >   s.commission
                > FROM customer c
                > JOIN salesman s
                >   ON c.salesman_id = s.salesman_id
                > WHERE s.commission > 0.12
                >   AND c.city <> s.city;
```

Please submit sql's for above five questions as your assignment answers.

**********Section 2**********

1. Please complete the below create table commands on complex data and create the table in hive.

Replace ? with appropriate delimiter according to the given data in below create table sql

```
salesdetail_complex (Product_ID INT,productdetails map<String,String>,Order_Priority VARCHAR(4),merchantType CHAR(4),Sale_Amount DOUBLE,Order_Quantity BIGINT,Discount FLOAT,Salaryhike TINYINT,companyprofit SMALLINT, financeDeficit DECIMAL(8,2),indian BOOLEAN,saledate array<date>,saleyear array<int>,selleramountfile array<DOUBLE>,orderQuantityfile array<BIGINT>,costlist map<int,int>,strutureType struct<city:string,state:string,pin:bigint>,systemdatetime array<String>) ROW FORMAT DELIMITED FIELDS TERMINATED BY '?' collection items terminated by '?' map keys terminated by '?';
```

And load below sample data in above table

```
1,Cam1eras#Cameras$Cameras#Cameras,Medium,Seller,750000,5000000,1500.24,100,1000,1500.659744,TRUE,2012-12-21$2013-12-26,2012$1998,750700.00$850000.01,500000065$500056458,401#901$1200#5410,ap$mp$500001,2019-12-21$2020-12-26 12:00
2,Cam`eras#Nikon$DLR#SmileDetector,Not Specified,Dealer,750001,5000561,1501.24,101,1001,1501.659744,FALSE,2012-12-22$2012-12-27,2012$1999,750001.00$750000.02,500056187$500056458,,ap$mp$500002,2019-12-21$2020-12-26 12:01
3,,High,Seller,750002,5000562,1502.24,105,1002,1502.659744,FALSE,2012-12-23$2012-12-28,2012$2000,750002.00$780000.03,500056245$500056458,400#902$1200#5413,ap$mp$500003,2019-12-21$2020-12-26 12:02
4,Accessories#PentaLite$TV Accessories#Wall Mount,Critical,Dealer,750003,5000563,1503.24,106,1003,1503.659744,FALSE,2012-12-24$2012-12-29,2012$2001,750003.00$790000.04,500056345$500056458,400#903$1200#5414,ap$mp$500004,2019-12-21$2020-12-26 12:03
5,Camera1s#Timbre$Video#Lens Kit,Low,Dealer,750004,5000564,1504.24,108,1004,1504.659744,FALSE,2012-12-25$2012-12-30,2012$2002,750004.00$800000.05,500056458$500056458,400#904$1200#5415,ap$mp$500005,2019-12-21$2020-12-26 12:04
```

2. Write a sql to get the only 2 records from salesdetail_complex

```
hive (hive_test)> SELECT * FROM salesdetail_complex LIMIT 2;
OK
1       {"Cam1eras":"Cameras","Cameras":"Cameras"}      Medium  Seller  750000.0        5000000 1500
.24     100     1000    1500.66 true    ["2012-12-21","2013-12-26"]     [2012,1998]     [750700.0,85
0000.01]        [500000065,500056458]   {401:901,1200:5410}     {"city":"ap","state":"mp","pin":5000
01}     ["2019-12-21","2020-12-26 12:00"]
2       {"Cam`eras":"Nikon","DLR":"SmileDetector"}      Not Specified   Dealer  750001.0        5000
561     1501.24 101     1001    1501.66 false   ["2012-12-22","2012-12-27"]     [2012,1999]     [750
001.0,750000.02]        [500056187,500056458]   {}      {"city":"ap","state":"mp","pin":500002} ["20
19-12-21","2020-12-26 12:01"]
Time taken: 0.088 seconds, Fetched: 2 row(s)
```

3. Create a non **partitioned table** named as **non_part** on below data.
Table metadata for non partitioned table:

Table columns

```
dateid smallint ,
caldate date ,
day string ,
week smallint ,
month string ,
qtr string ,
year smallint ,
holiday boolean
```

Sample data from above table.

```
1827|2008-01-01|WE|1|JAN|1|2008|TRUE
1828|2008-01-02|TH|1|JAN|1|2008|FALSE
1829|2008-01-03|FR|1|JAN|1|2008|FALSE
1830|2008-01-04|SA|2|JAN|1|2008|FALSE
1831|2008-01-05|SU|2|JAN|1|2008|FALSE
1832|2008-01-06|MO|2|JAN|1|2008|FALSE
1833|2008-01-07|TU|2|JAN|1|2008|FALSE
1834|2008-01-08|WE|2|JAN|1|2008|FALSE
1835|2008-01-09|TH|2|JAN|1|2008|FALSE
1836|2008-01-10|FR|2|JAN|1|2008|FALSE
1837|2008-01-11|SA|3|JAN|1|2008|FALSE
1838|2008-01-12|SU|3|JAN|1|2008|FALSE
1839|2008-01-13|MO|3|JAN|1|2008|FALSE
1840|2008-01-14|TU|3|JAN|1|2008|FALSE
1841|2008-01-15|WE|3|JAN|1|2008|FALSE
1842|2008-01-16|TH|3|JAN|1|2008|FALSE
1843|2008-01-17|FR|3|JAN|1|2008|FALSE
1844|2008-01-18|SA|4|JAN|1|2008|FALSE
1845|2008-01-19|SU|4|JAN|1|2008|FALSE
1846|2008-01-20|MO|4|JAN|1|2008|FALSE
1847|2008-01-21|TU|4|JAN|1|2008|FALSE
1848|2008-01-22|WE|4|JAN|1|2008|FALSE
1849|2008-01-23|TH|4|JAN|1|2008|FALSE
1850|2008-01-24|FR|4|JAN|1|2008|FALSE
1851|2008-01-25|SA|5|JAN|1|2008|FALSE
1852|2008-01-26|SU|5|JAN|1|2008|FALSE
```

```
hive (hive_test)> CREATE TABLE non_part (
                >   dateid SMALLINT,
                >   caldate DATE,
                >   day STRING,
                >   week SMALLINT,
                >   month STRING,
                >   qtr STRING,
                >   year SMALLINT,
                >   holiday BOOLEAN
                > )
                > ROW FORMAT DELIMITED
                > FIELDS TERMINATED BY '|'
                > STORED AS TEXTFILE;
OK
Time taken: 0.062 seconds
hive (hive_test)> LOAD DATA INPATH '/data/hive/nonpart/non_part.txt'
                > INTO TABLE non_part;
Loading data to table hive_test.non_part
OK
Time taken: 0.111 seconds
hive (hive_test)> SELECT * FROM non_part LIMIT 5;
OK
1827    2008-01-01      WE      1       JAN     1       2008    true
1828    2008-01-02      TH      1       JAN     1       2008    false
1829    2008-01-03      FR      1       JAN     1       2008    false
1830    2008-01-04      SA      2       JAN     1       2008    false
1831    2008-01-05      SU      2       JAN     1       2008    false
Time taken: 0.088 seconds, Fetched: 5 row(s)
```

4. Now create a partitioned table on column caldate named as date_part from non partitioned table created above

Partitioned table metadata

Table Columns

```
dateid smallint,
 day string ,
 week smallint ,
 month string ,
 qtr string ,
 year smallint ,
 holiday boolean
```

Partition column

```
caldate
```

```
hive (hive_test)> SHOW PARTITIONS date_part;
OK
caldate=2008-01-01
caldate=2008-01-02
caldate=2008-01-03
caldate=2008-01-04
caldate=2008-01-05
caldate=2008-01-06
caldate=2008-01-07
caldate=2008-01-08
caldate=2008-01-09
caldate=2008-01-10
caldate=2008-01-11
caldate=2008-01-12
caldate=2008-01-13
caldate=2008-01-14
caldate=2008-01-15
caldate=2008-01-16
caldate=2008-01-17
caldate=2008-01-18
caldate=2008-01-19
caldate=2008-01-20
caldate=2008-01-21
caldate=2008-01-22
caldate=2008-01-23
caldate=2008-01-24
caldate=2008-01-25
caldate=2008-01-26
Time taken: 0.076 seconds, Fetched: 26 row(s)
```

5. Now create a partitioned and bucketed table on above table

Table Columns

```
dateid smallint,
week smallint ,
month string ,
qtr string ,
year smallint ,
holiday boolean
```

Partition column

```
caldate
```

Bucket column

```
dateid
```

```
hive (hive_test)> SHOW PARTITIONS date_part_bucket;
OK
caldate=2008-01-01
caldate=2008-01-02
caldate=2008-01-03
caldate=2008-01-04
caldate=2008-01-05
caldate=2008-01-06
caldate=2008-01-07
caldate=2008-01-08
caldate=2008-01-09
caldate=2008-01-10
caldate=2008-01-11
caldate=2008-01-12
caldate=2008-01-13
caldate=2008-01-14
caldate=2008-01-15
caldate=2008-01-16
caldate=2008-01-17
caldate=2008-01-18
caldate=2008-01-19
caldate=2008-01-20
caldate=2008-01-21
caldate=2008-01-22
caldate=2008-01-23
caldate=2008-01-24
caldate=2008-01-25
caldate=2008-01-26
Time taken: 0.058 seconds, Fetched: 26 row(s)
```

```
hive (hive_test)> SELECT * FROM date_part_bucket WHERE caldate='2008-01-01';
OK
1827    1       JAN     1       2008    true    2008-01-01
Time taken: 0.103 seconds, Fetched: 1 row(s)
hive (hive_test)>
```